

Fanyou Wu | 吴凡优

Applied Scientist, Tech Lead · Amazon, PXT ITDS · Seattle, WA

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SUMMARY

Applied Scientist and tech lead building production LLM systems at Amazon, spanning LLM agents, reinforcement learning, synthetic data, and the AI-native software development lifecycle (SDLC). Author of 25 papers at venues including ACL, EMNLP, COLM, and AACL-IJCNLP. Winner or runner-up in 9 international ML competitions (KDD Cup, NeurIPS, CVPR, IJCAI), with about US\$62,000 in prize money.

EXPERIENCE

Applied Scientist, Tech Lead — Amazon, PXT ITDS Seattle, WA · Jan 2023 – present

- **AI-native SDLC (2026):** lead the team's move to an AI-native software development lifecycle, with LLM agents taking part in design, coding, testing, and deployment.
- **HR chatbot (2026):** build an internal HR chatbot that combines retrieval-augmented generation (RAG) with an LLM agent to answer policy questions and automate multi-step HR workflows.
- **Conversation Coach (2025):** develop a voice-enabled role-playing agent that simulates realistic employee conversations so HR professionals can practice difficult discussions; published at EMNLP 2026 Industry Track.
- **Job recommendation (2023–2024):** build a transformer-based job recommender on large-scale click-stream data, improving candidate-to-role matching across Amazon's hiring funnel.
- **Research (throughout):** publish on dialog generation, hallucination, fairness, unlearning, tool-augmented reasoning, and role playing at ACL, EMNLP, COLM, and AACL-IJCNLP.

Postdoctoral Researcher — Purdue University, Forestry and Natural Resources West Lafayette, IN · Jan 2022 – Dec 2022

- Built an iOS app with ARKit and Core ML that estimates tree diameter, height, and species on device.
- Applied PyTorch semantic segmentation to detect growth-ring edges in highly noisy wood images, released with a new annotated dataset.

Graduate Research Assistant — Purdue University West Lafayette, IN · Jan 2018 – Dec 2021

- Built deep learning systems for wood and bark species identification, including a portable knowledge-distilled model (Deep BarkID).
- Co-developed reinforcement learning and spatiotemporal models for ride-hailing dispatch, traffic forecasting, and last-mile routing.

Machine Learning Engineer — Envision Digital Shanghai, China · Aug 2017 – Dec 2017

- Forecast time-series electricity demand with LightGBM on a Hadoop/Hive-based IoT platform.

EDUCATION

Ph.D. in Forestry, Purdue University Jan 2018 – Dec 2021

M.S. in Wood Material, University of Eastern Finland Sep 2015 – Apr 2017

Exchange student, Wood Products Processing, University of British Columbia Aug 2013 – May 2014

B.E. in Wood Material, Nanjing Forestry University Sep 2011 – Jun 2015

SKILLS

- **LLMs and agents:** RAG, LLM agents and tool use, voice agents, synthetic data generation, hallucination and safety, fairness and unlearning, AI-native SDLC.
- **Machine learning:** reinforcement learning (Deep Q-learning, dispatching, repositioning), transformers and recommender systems, computer vision (segmentation, classification, knowledge distillation), spatiotemporal forecasting, gradient boosting (LightGBM).
- **Engineering:** Python, PyTorch, Julia, Swift (ARKit, Core ML), Hadoop/Hive, distributed training, TypeScript/Next.js.

COMPETITIONS

13 international ML competitions: 4 first-place and 5 second-place finishes, about US\$62,300 in prizes.

Year	Competition	Result	Prize (US\$)	Solution
2024	KDD Cup 2024: Meta Comprehensive RAG Benchmark (CRAG)	Category winner in 2 tracks	1,000	
2022	KDD Cup 2022: Amazon ESCI Product Search	2nd in Tasks 2 and 3	4,000	Code
2021	CVPR 2021: 2nd Agriculture-Vision Prize Challenge	2nd in 2 tracks	4,200	Code
2021	Amazon Last Mile Routing Research Challenge	6th	—	Code
2021	KDD Cup 2021: City Brain Challenge (RL Track)	10th	500	Code
2020	KDD Cup 2020: Reinforcement Learning Track	1st in Reposition track	8,000	Report
2020	NeurIPS 2020: Traffic4cast	2nd	5,000	Code
2019	IJCAI-19: Alibaba Adversarial AI Challenge	1st in Targeted Attack track	5,000	Code
2019	KDD Cup 2019: Regular ML Track	4th	1,000	
2018	JDD 2018 Global Digitalization Challenge	2nd in final; 1st in North America	~23,000	
2018	Alibaba Cloud 3rd Annual Security Algorithm Challenge	8th	~1,500	
2017	Dahang Cup	2nd	~7,600	
2017	Alibaba Cloud 2nd Annual Security Algorithm Challenge	1st in final (~6th overall)	~1,500	

PUBLICATIONS

25 papers, 11 as first author. Also on [Google Scholar](#).

2026

1. **F. Wu**, S. Maharjan, A. Yessenalina, D. X. Chen, R. Srivastava, S. H. Sengamedu. Conversation Coach: A Voice-enabled AI System that Helps Practice Difficult Workplace Conversations. *EMNLP 2026, Industry Track*. [arXiv:2609.00441](#)
2. B. K. Baghel, **F. Wu**, S. Maharjan, P. Yao, S. H. Sengamedu. GEAR: Training-Free Rule Distillation for Advanced and Efficient Tool-Augmented Reasoning. *AACL-IJCNLP 2026*. [OpenReview](#)

2025

3. Z. Liu, S. Maharjan, **F. Wu**, R. Parikh, B. Bayar, S. H. Sengamedu, M. Jiang. Disentangling Biased Knowledge from Reasoning in Large Language Models via Machine Unlearning. *ACL 2025*. [Paper](#)
4. Z. Zhang, W. Xu, **F. Wu**, C. K. Reddy. FalseReject: A Resource for Improving Contextual Safety and Mitigating Over-Refusals in LLMs via Structured Reasoning. *COLM 2025*. [arXiv:2505.08054](#)
5. G. Yeo, F. Tan An Ting, K. Jaidka, S. Furniturewala, **F. Wu**, W. Xu, V. Jain, A. Chadha, Y. Liu, S.-K. Ng. PHAnToM: Persona-Based Prompting Has an Effect on Theory-of-Mind Reasoning in Large Language Models. *ICWSM 2025*. [Paper](#)
6. C. Warner, **F. Wu**, R. Gazo, B. Benes, S. Fei. Environmental Sensitivity in AI Tree Bark Detection: Identifying Key Factors for Improving Classification Accuracy. *Algorithms 18(7)*, 2025. [Paper](#)

2024

7. **F. Wu**, W. Xu, C. K. Reddy, S. H. Sengamedu. Synthesizing Conversations from Unlabeled Documents using Automatic Response Segmentation. *Findings of ACL 2024*. [Paper](#)
8. W. Xu, J. Desai, **F. Wu**, J. Valvoda, S. H. Sengamedu. HR-Agent: A Task-Oriented Dialogue (TOD) LLM Agent Tailored for HR Applications. *arXiv 2024*. [arXiv:2410.11239](#)
9. C. Warner, **F. Wu**, R. Gazo, B. Benes, N. Kong, S. Fei. CentralBark Image Dataset and Tree Species Classification Using Deep Learning. *Algorithms 17(5)*, 2024. [Paper](#)
10. **F. Wu**, Y. Huang, B. Benes, C. C. Warner, R. Gazo. Automated Tree Ring Detection of Common Indiana Hardwood Species through Deep Learning: Introducing a New Dataset of Annotated Images. *Information Processing in Agriculture 11(4):552–558*, 2024. [Paper](#)

2023

11. W. Xu, W. Hu, **F. Wu**, S. H. Sengamedu. DeTiME: Diffusion-Enhanced Topic Modeling using Encoder-decoder based LLM. *Findings of EMNLP 2023*.
12. Y. Wan, **F. Wu**, W. Xu, S. H. Sengamedu. Sequence-Level Certainty Reduces Hallucination in Knowledge-Grounded Dialogue Generation. *arXiv 2023*. [arXiv:2310.18794](#)
13. Y. Liu, **F. Wu**, Z. Liu, K. Wang, F. Wang, X. Qu. Can Language Models Be Used for Real-World Urban-Delivery Route Optimization? *The Innovation*, 2023. [Paper](#)

2022

14. **F. Wu**, Y. Liu, R. Gazo, B. Benes, X. Qu. Some Practice for Improving the Search Results of E-commerce. *KDD Cup 2022 Workshop*. [arXiv:2208.00108](#)

15. **F. Wu**, C. Lyu, Y. Liu. A Personalized Recommendation System for Multi-modal Transportation Systems. *Multimodal Transportation* 1(2):100016, 2022. [Paper](#)
16. Y. Liu, **F. Wu**, C. Lyu, S. Li, J. Ye, X. Qu. Deep Dispatching: A Deep Reinforcement Learning Approach for Vehicle Dispatching on Online Ride-Hailing Platform. *Transportation Research Part E*, 2022. [Paper](#)

2021

17. **F. Wu**, R. Gazo, B. Benes, E. Haviarova. Deep BarkID: A Portable Tree Bark Identification System by Knowledge Distillation. *European Journal of Forest Research*, 2021. [Paper](#)
18. **F. Wu**, R. Gazo, E. Haviarova, B. Benes. Wood Identification Based on Longitudinal Section Images by Using Deep Learning. *Wood Science and Technology* 55(2):553–563, 2021. [Paper](#)
19. Y. Liu, **F. Wu**, C. Lyu, X. Liu, Z. Liu. Behavior2Vector: Embedding Users' Personalized Travel Behavior to Vector. *IEEE Transactions on Intelligent Transportation Systems*, 2021. [Paper](#)
20. **F. Wu**, Y. Liu. A Hybrid Method with Rules and Optimization to Solve the Last-Mile Delivery Problems. *Technical Proceedings of the Amazon Last Mile Routing Research Challenge*, 2021. [Paper](#)
21. M. Kopp, D. Kreil, M. Neun, ..., **F. Wu**, Y. Liu, ..., S. Hochreiter. Traffic4cast at NeurIPS 2020: Yet More on the Unreasonable Effectiveness of Gridded Geo-Spatial Processes. *NeurIPS 2020 Competition and Demonstration Track*, PMLR 133:325–343, 2021. [Paper](#)

2020 and earlier

22. **F. Wu**, Y. Liu, Z. Liu, X. Qu, R. Gazo, E. Haviarova. TLab: Traffic Map Movie Forecasting Based on HR-NET. *arXiv 2020*. [arXiv:2011.07728](#)
23. Y. Liu, **F. Wu**, B. Yu, Z. Liu, J. Ye. Building Effective Large-Scale Traffic State Prediction System: Traffic4cast Challenge Solution. *arXiv 2019*. [arXiv:1911.05699](#)
24. **F. Wu**, R. Gazo, E. Haviarova, B. Benes. Efficient Project Gradient Descent for Ensemble Adversarial Attack. *IJCAI-19 AIBS Workshop*. [arXiv:1906.03333](#)
25. **F. Wu**, P. P. Kärenlampi. Phase Transition in a Growing Network. *Journal of Complex Networks* 6(5):788–799, 2017. [Paper](#)

INVITED TALKS

- Nov 2021 Some Machine Learning Applications about Tree and Wood ([slides](#)). Digital Forestry Retreat.
- Jun 2021 Team WRL: Solution for CVPR 2021 Agriculture-Vision Prize Challenge ([video](#)). CVPR 2021 Agriculture-Vision Workshop.
- Dec 2020 TLab: Traffic Map Movie Forecasting Based on HR-NET. NeurIPS 2020 Traffic4cast Special Session.
- Dec 2020 Some Experience about Machine Learning Competitions: KDD Cup RL Track as an Example. SAMS Transportation Virtual Workshop.
- Oct 2020 KDD Cup 2020 RL Track Winners Presentation, Part II. Reinforcement Learning Algorithm & Application Virtual Seminar.

POSTERS

- Applying Zero-Shot Learning to Wood Identification. SWST.
- Efficient Project Gradient Descent for Ensemble Adversarial Attack. IJCAI-19 AIBS Workshop.
- Deep Learning for Lumber Identification. ISHCP.
- A Mobile App for Common Midwestern Hardwood Species Identification. Purdue FNR Research Poster Competition.

OPEN-SOURCE CONTRIBUTIONS

- [KDD-Cup-2022-Amazon](#): NLP retrieval and ranking models for e-commerce search.
- [WRL-Agriculture-Vision](#): farmland semantic segmentation with distributed training and model ensembles.
- [Traffic4Cast-2020-TLab](#): HR-Net spatiotemporal traffic forecasting.
- [TLab-City-Brain-Challenge](#): reinforcement learning for city-scale traffic signal control.
- [TLab-Last-Mile](#): rules plus optimization for last-mile delivery routing.
- [EPGD](#): efficient projected gradient descent for ensemble adversarial attacks.
- [DBID](#) and [lumber_identification](#): wood and bark species identification models.

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